

Lavanya Talapaneni

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PROFESSIONAL SUMMARY

- Around 6+ years of experience in data engineering, building scalable and high-performance data pipelines.
- Strong in Python, SQL, and PySpark for data transformation, automation, and workflow orchestration.
- Good experience in Amazon Web Services like S3, IAM, EC2, EMR, Kinesis, VPC, Dynamo DB, RedShift, Amazon RDS, Lambda, Athena, Glue, DMS, Quick Sight, Amazon Elastic Load Balancing, Auto Scaling, CloudWatch, SNS, SQS and other services of the AWS family.
- Hands-on expertise with AWS services (S3, Glue, Lambda, CloudWatch, Step Functions, EMR) for cloud data integration.
- Implement NiFi to Spark streaming directly without using Kafka internally to provide various options to client in single Confidential.
- Monitoring tasks using bash scripts such as monitoring ASM disk space, standby database log repots, tablespace report.
- Experience in building power bi reports on Azure Analysis services for better performance.
- Deep experience with Snowflake, Redshift, and SQL Server for warehousing and analytics.
- Proficient in handling structured, semi-structured (JSON, Parquet, Avro), and unstructured data.
- Strong knowledge of data modeling techniques (Star/Snowflake Schema, SCDs, partitioning).
- Experienced in big data processing with Spark, Hive, and Hadoop ecosystems.
- Adept in DevOps practices: CI/CD pipelines with Git, Jenkins, Docker, and Terraform.
- Exposure to data governance, lineage, and quality checks to ensure reliable datasets.
- Skilled in ETL/ELT design using Talend, AWS Glue, and Apache Airflow for batch and streaming workloads.
- Experience working with NoSQL databases like Cassandra and HBase and developed real-time read/write access to very large datasets via HBase.
- Developed Spark Applications that can handle data from various RDBMS (MySQL, Oracle Database) and Streaming sources.
- Designing and deployment of Hadoop cluster and different Big Data analytic tools including Apache Spark, with Cloudera Distribution.
- Worked on different file formats like JSON, XML, CSV, ORC, Parquet. Experience in processing both structured and semi structured Data with the given file formats.
- Experience in developing Web Services like SOAP, REST, Restful with Python programming language.
- Experience in moving data between GCP and Azure using Azure Data Factory.
- Having good experience in spark core, spark SQL and spark streaming.
- Developed Spark to load the database log files from Linux file system to the HDFS for further processing.
- Implement Spark Kafka streaming to pick up the data from Kafka and send to Spark pipeline.
- Proven knowledge of standards-compliant, cross-browser compatible HTML, CSS, JavaScript, and Ajax.
- Delivered business insights by enabling data visualization in Power BI, Tableau, and Amazon QuickSight.
- Proven ability to collaborate with cross-functional teams and deliver end-to-end data solutions

that reduce cost and improve performance.

Technical Skills

- **Programming & Scripting:** Python, SQL, PySpark, Shell Scripting, Scala (basic), Java (basic)
- **ETL & Data Integration:** Talend Cloud, AWS Glue, Apache Airflow, Apache NiFi, Kafka, Informatica
- **Data Warehousing & Databases:** Snowflake, Amazon Redshift, SQL Server, MySQL, PostgreSQL, Oracle
- **Big Data & Processing Frameworks:** Hadoop, Spark (PySpark/Scala), Hive, Presto, Apache Flink
- **Cloud Platforms & Services:**
 - **AWS:** S3, Glue, Lambda, CloudWatch, Step Functions, EMR, Athena, Kinesis, IAM
 - **Azure:** Data Factory, Synapse Analytics, Databricks
 - **GCP:** BigQuery, Cloud Storage, Dataflow
- **Data Modeling & Storage:** Star/Snowflake Schema, Dimensional Modeling, Slowly Changing Dimensions (SCD Types 1–3), Partitioning, Clustering, Indexing
- **Data Formats:** JSON, Avro, ORC, Parquet, CSV, XML, Delta Lake
- **Version Control & CI/CD:** Git, GitHub, GitLab, Jenkins, Docker, Kubernetes, Terraform, Ansible
- **Data Governance & Quality:** Data Lineage, Data Catalog, Great Expectations, Deequ, Data Validation Frameworks
- **Workflow & Orchestration:** Apache Airflow, AWS Step Functions, Control-M, Luigi
- **Monitoring & Logging:** AWS CloudWatch, Splunk, ELK Stack, Datadog
- **Visualization & BI Tools:** Power BI, Tableau, Amazon QuickSight

Professional Experience

Truist Bank
Data Engineer, Dallas, TX

Nov 2024 – Present

Responsibilities:

- Designed and optimized metadata-driven ETL/ELT pipelines in Snowflake and AWS.
- Developed incremental and batch ingestion frameworks for S3 → Snowflake → SQL Server flows.
- Implemented restart flag validation and dynamic JSON metadata handling to enable flexible ingestion.
- Designed and Co-ordinated with team in implementing Advanced Analytical Models in Hadoop Cluster over large Datasets
- Programming Tools and writing scripts in various operating systems like Terminal, Bash and PowerShell Mac, Linux and Windows.
- Experience in building Azure pipelines using the Microsoft agent and running python file related tests.
- Built automated monitoring and alerting using AWS CloudWatch, Step Functions, and Lambda.
- Created partitioning and clustering strategies in Snowflake to reduce query cost and runtime.
- Integrated semi-structured formats (JSON, Parquet, Avro) into enterprise data warehouse.
- Deployed pipelines with CI/CD automation (Git, Jenkins, Docker) for consistency.
- Used AWS Glue for transformations and AWS Lambda to automate the process.
- Built role-based access control and ensured compliance with enterprise data governance standards.

- Used DMS to migrate tables from homogeneous and heterogeneous DBs from On-premise to AWS Cloud.
- Configured Spark streaming to receive real time data from the Kafka and store the stream data to HDFS using Scala.
- Created an internal tool to compare RDBMS and Hadoop, ensuring that all data in the source and target matches using a shell script, reducing the complexity of data movement.
- Designed Pipelines with Apache Beam, KubeFlow, Dataflow and orchestrated jobs into GCP.
- Worked on GCP POC to migrate data and applications from On-Prem to Google Cloud.
- Worked closely with analytics teams to publish curated datasets for BI platforms.
- Implemented real-time data streaming pipeline using AWS Kinesis, Lambda, and Dynamo DB and deployed AWS Lambda code from Amazon S3 buckets.
- Used Amazon Elastic Kubernetes Service (Amazon EKS) to run and scale Kubernetes applications in the cloud or on-premises.
- Experience in created Azure Data Factory pipelines for migrating the on-premises data from and Azure Sql Data Warehouse.
- Wrote various data normalization jobs for new data ingested into Redshift.
- Reduced pipeline failures by implementing end-to-end logging and retry mechanisms.
- Supporting continuous storage in AWS using Elastic Block Storage, S3 and created Volumes and configured Snapshots for EC2 instances.
- Developed data pipeline using Flume, python map reduce to ingest various sources data into HDFS for analysis.
- AWS SQS was used to transfer the processed data to the next working teams for processing.
- Developed a data pipeline using Kafka and Storm to store data into HDFS.
- Developed bash scripts for importing the Oracle database tables to HDFS.
- Integrated data from MySQL to the Client Portal using AWS Lambda services.

Cigna Health Care
Data Engineer, Texas

Aug 2023 to Aug 2024

Responsibilities:

- Collaborated with the client to understand their requirements and formulated a detailed design plan in conjunction with the team.
- Engaged with the client team to validate the design and made necessary modifications based on their feedback and evolving requirements.
- Extracted data from DB2 and transferred it to AWS for subsequent analysis, visualization, and report generation.
- Exceptional skills in Automating repetitive tasks using Python, UNIX Bash Shell Scripting.
- Responsible for building scalable distributed data solutions using Big Data technologies like Apache Hadoop, MapReduce, Shell Scripting, Hive.
- Experience in automation of code deployment, support, and administrative tasks across multiple cloud providers such as AWS, Microsoft Azure, GCP.
- Established HBase tables and configured columns to efficiently store user event data.
- Utilized Hive and Impala for querying the data stored in HBase, ensuring efficient retrieval and analysis.
- Developed and deployed core API services using Scala and Spark, enabling seamless interaction with the data.
- Migrated on premise database structure to Confidential Redshift data warehouse
- Leveraged Spark's capabilities to manage data frames, perform migrations between AWS and MySQL, and execute complex ETL processes.

- Implemented a continuous ETL pipeline using Kafka, Spark Streaming, and HDFS, ensuring smooth data flow and processing.
- Configured Qlik Replicate to streamline data integration between healthcare systems, supporting data-driven decision-making for patient care and operational efficiency.
- Designed replication tasks to handle complex data structures specific to healthcare, including claims data, provider details, and patient records, ensuring compliance with HIPAA regulations.
- Implemented robust data transformation rules to standardize formats across varied healthcare data sources, improving data usability for reporting and analytics.
- Experience on Azure Databricks cloud to organizing the data into notebooks and making it easy to visualize data through the use of dashboards.
- Established secure, encrypted data pipelines in Qlik Replicate for transmitting sensitive patient information between systems, safeguarding against unauthorized access.
- Used the Bash Shell scripts to automate the deployment process.
- Experience with Linux Shell Scripting and in Python programming for database operations.
- Build data pipelines in airflow in GCP for ETL related jobs using different airflow operators.
- Conducted performance tuning on Qlik Replicate tasks for high-volume healthcare data transfers, minimizing latency and ensuring real-time data availability for critical care systems.
- Developed Qlik Replicate monitoring and alerting protocols tailored for healthcare applications, enabling proactive management of data workflows to prevent disruptions in patient services.

Acaira Technologies
Data Engineer, India

Aug2019 – Dec 2022

Responsibilities:

- Built and maintained ETL pipelines using Talend and SQL Server for enterprise reporting.
- Migrated on-prem SQL workflows into AWS S3 + Redshift, improving scalability.
- Automated ingestion from flat files, APIs, and relational DBs into cloud warehouses.
- Improved SQL query runtime by 30% through indexing, partitioning, and query tuning.
- Developed stored procedures and triggers for data transformations.
- Integrated Docker and Jenkins to automate CI/CD for ETL deployments.
- Monitored workflows with Talend Administration Console (TAC) and alerting.
- Partnered with BI teams to design star schemas and fact/dimension models.
- Built reports for monitoring data loads into GCP and drive reliability at the site level
- Work on building a framework for data processing on AWS Glue, to increase speed, efficiency and decrease costs.
- Experience in Linux Bash scripting and following Guidelines in Python.
- Written Storm topology to accept the events from Kafka producer and emit into Cassandra DB.
- Ingesting data from ORACLE RDBMS, Tera data systems into Hadoop data lake.
- Secured access to data pipelines with vault-based connection management.
- Designing the distribution strategy for tables in Azure SQL data warehouse
- Designed logging and error-handling framework for Talend jobs.
- Used AWS Glue for the data transformation, validate and data cleansing.

Jio, India
Data Engineer

May 2018 – May 2019

Responsibilities:

- Assisted in designing SQL-based ETL mappings and transformations for client projects.
- Created stored procedures, views, and queries to support reporting needs.
- Migrated flat-file ingestion workflows into SQL-based ETL processes.
- Built test datasets and reconciliation reports to validate ETL outputs.
- Data Modeling with PostgreSQL to design Fact and Dimension tables using Snowflake Methodologies
- Designed Database Management Console and Application Status page using PERL CGIs and regex, on Apache.
- Automating repeated tasks using Python and UNIX Bash Scripting.
- Worked with Senior Engineer on configuring Kafka for streaming data.
- Worked on Continuous Integration CI/CD pipeline for Azure Cloud Services
- Imported data from RDBMS to Hadoop import.
- Documented data dictionaries and ETL workflows for team knowledge base.
- Setting up S3 bucket and Access Control policies using IAM
- Supported data modeling activities: star schema design and normalization.
- Worked with semi-structured inputs (CSV, JSON) to standardize ingestion.
- Performed unit and integration testing of new ETL jobs.
- Gained exposure to orchestration tools and scheduling concepts.
- Presented progress updates and technical findings to senior engineers.

Education

Master of Science in Computer & Information Science
Texas A&M University-Commerce | Jan 2023 – Aug 2024

Certifications

- AWS Certified Data Engineer – Associate
- AWS Certified Solutions Architect – Associate
- AWS Certified Cloud Practitioner